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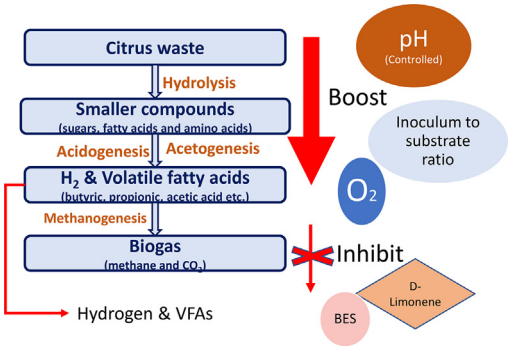
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Effect of pH, substrate loading, oxygen, and methanogens inhibitors on volatile fatty acid (VFA) production from citrus waste by anaerobic digestion

Bahriye Eryildiz[a,b], Lukitawesa[b,*], Mohammad J. Taherzadeh[b]

[a] Istanbul Technical University, Environmental Engineering Department, Maslak, 34469 Istanbul, Turkey
[b] Swedish Centre for Resource Recovery, University of Borås, SE 501-90 Borås, Sweden

GRAPHICAL ABSTRACT



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ABSTRACT

Citrus waste from e.g., juice production is a potential substrate for anaerobic digestion (AD). However, due to the toxic citrus peel oil content, citrus waste has several challenges in biogas production. Hence, volatile fatty acids (VFAs) are very interesting intermediate products of AD. This paper was aimed to investigate VFA production from citrus wastes by boosting its production and inhibiting methane formation. Therefore, the effects of inoculum to substrate ratio (ISR), O_2 presence, pH, and inhibitor for methanogens, in VFA production from citrus waste through acidification process were studied. The addition of 2 g/L methanogens inhibitor and the presence of O_2 in the reactors were able to reduce methane production. The highest yield of VFA (0.793 g VFA/g VS_{added}) was achieved at controlled pH at 6 and low substrate loading (ISR 1:1). Acetic acid (32%), caproic acid (21%), and butyric acid (15%) dominate the VFA composition in this condition.

1. Introduction

Citrus is one of the most abundant genus of fruits in the world. According to the Food and Agriculture Organization of the United Nations (FAO), global citrus production increased by 20.6% from 2006 to 2016, to 132.4 million tones (FAO, 2017). Approximately, 98% of the industrialized citrus harvest consists of orange, mandarin, grapefruit, and lemon. Citrus fruits are mainly processed to obtain fruit juice. They are also used in many fields such as the canning industry, chemical industry, food industry, cosmetics, and pharmaceutical industries

* Corresponding author.
E-mail address: Lukitawesa.Lukitawesa@hb.se (Lukitawesa).

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(Marín et al., 2007; Wu et al., 2017). Approximately, 40–60% of citrus production is used in fruit juice production, and as a result of this process, 50–60% of the citrus mass ends up in waste streams (Wikandari et al., 2014a). Dumping and landfilling for citrus waste causes several environmental challenges such as unpleasant odor and leachates. This wastes attracts flies and rats due to its low pH, high water content, and high organic matter content, and high biodegradability (Ruiz and Flotats, 2014; Wikandari et al., 2014a). Therefore, alternative methods to face these challenges have been investigated. The high moisture content of citrus wastes is a limitation and make it unsuitable for combustion and incineration. The possibility of energy production by anaerobic digestion which is more environmentally friendly than other alternatives (Xu et al., 2018).

Anaerobic digestion is a biological process that can be used for treating organic wastes. The digestion performance (e.g. biogas production, organic reduction, inhibition characteristics, and process stability) is highly dependent on the feedstock's characteristics (Li et al., 2017). The anaerobic process consists of four main stages: hydrolysis, acidogenesis, acetogenesis, and methanogenesis (Wainaina et al., 2019). In the first phase of the anaerobic digestion process, i.e. hydrolysis, complex organic polymers are broken down into simple soluble monomers by means of extracellular enzymes formed by hydrolytic microorganisms (Debruyn and Hilborn, 2007). During the acidogenesis phase, products formed in the hydrolysis phase are converted to volatile fatty acids (VFAs) and minor products such as hydrogen, carbon dioxide (CO_2), and acetic acid by acidogenic bacteria. In the acetogenesis phase, acetogenic bacteria further convert VFAs to acetate, CO_2, and/or hydrogen (Debruyn and Hilborn, 2007; Khan et al., 2016). In the last phase, methanogenic microorganisms convert products of acidogenesis and acetogenesis into carbon dioxide and methane (Massé et al., 2011). However, citrus waste has been known as a difficult substrate for methane production due to the citrus peel oil content which is toxic for the methanogens (Fuertes, 2015; Kurniawan et al., 2018; Lukitawesa et al., 2018; Ruiz et al., 2016; Ruiz and Flotats, 2014; Ruiz and Flotats, 2016; Wikandari et al., 2014a; Wikandari et al., 2015). Citrus peel oil contains the monoterpenes limonene (65–77%), γ-terpinene (13–21%), α-pinene (1.5–3.0%), β-pinene (1.3–2.5%), β-myrcene (1.5–2.5%). There are also alcohols citronellol, geraniol, and linalool (combined total 1–1.5%) and smaller amounts of aldehydes, esters, and phenols (Clarke, 2008) present in the peel oil.

Recent studies have shown that the anaerobic digestion (AD) process could be designed to produce biogas as well as volatile fatty acids (Wainaina et al., 2019; Yin et al., 2016; Zhou et al., 2018). VFAs, which are a product of the acidogenesis phase, consist mainly of acetic acid, propionic acid, butyric acid, and valeric acid (Wainaina et al., 2019). The operational parameters, substrate composition, available microbial community, the process design, and type of reactor affect the amount and composition of VFAs produced (Li et al., 2013; Stein et al., 2017; Wainaina et al., 2019; Wang et al., 2014).

In order to produce VFAs, the methanogens that utilize VFAs should be inhibited. Several studies have shown that 2-bromoethanesulfonate (BES) has the ability to inhibit the activity of the methanogens (Chidthaisong and Conrad, 2000; Zinder et al., 1984); having an optimum concentration of around 50 Mm for complete inhibition (Zinder et al., 1984). Another study has reported that the addition of 1 μmol/mL of BES could reduce 60% of methanogen activity, leading to an increased acetate accumulation (Mohan et al., 2008). The previous study had been investigating the effect of different factors such as O_2 presence, inoculum to substrate ratio (ISR), BES, and pH towards VFA production from food waste (Lukitawesa et al., 2020). There is a scarce of study to investigate the effect of those factors at the same time for toxic feedstock, such as citrus waste.

In the form of VFA, acetic acid has the highest in the market size, with 3,500,000 tons/year and price per ton 800 $. It is followed by propionic acid with 180,000 tons/year and price per ton 1700 $. Butyric acid and caproic acid have the smallest market size at are 30,000 tons/year and 25,000 tons/year, respectively. The price of tons per ton is 2500 $ (Jankowska et al., 2017).

Volatile fatty acids (VFAs) are used in many fields such as bioplastics synthesis, biological nitrogen removal, and bioenergy production, for example, as alternatives to traditional external carbon sources in municipal wastewater treatment (Xie et al., 2014; Yin et al., 2016). In addition, with an appropriate method, VFA can be converted to alcohols, biohydrogen, microalgal lipids, bioelectricity, aldehydes, ketones, alkanes, polyhydroxyalkanoates (PHA), a biopolymer used for biodegradable plastics production, and can also find a direct usage as food additives in the food and the pharmaceutical industries (Dahiya et al., 2015; Khan et al., 2016; Trad et al., 2015; Wainaina et al., 2019).

The VFA production from citrus waste was previously reported in two-stage anaerobic digestion with the highest yield of 0.125 g VFA/g VS_{added} (Lukitawesa et al., 2018). The VFA concentration in a one-stage reactor which digests citrus waste at organic loading rate 3 g VS/L.d without pH control was around 6 g/L, which means the yield was only 0.072 g VFA/g VS_{added} (Wikandari et al., 2014a). However, those studies were mainly focused on producing methane instead of VFA. There is no information on whether citrus peel oil can also inhibit the other steps in anaerobic digestion i.e., hydrolysis and acidification process.

The objective of this study was to investigate the effect of various factors such as pH, BES, inoculum to substrate ratio, and oxygen on the production of VFA from citrus waste under anaerobic conditions. This study was divided into two cultures. In the first culture, the effect of BES, oxygen, and inoculum to substrate ratio on the concentration, yield, and composition of VFA produced from citrus waste was investigated. In the second culture, the effect of pH and inoculum to substrate ratio on VFA concentration, yield, and composition produced from citrus waste was investigated. Besides, biogas production as a byproduct and the compositions were investigated in both cultures.

2. Materials and method

2.1. Inoculum

Sludge used as inoculum was obtained from a digester treating food waste sludge and operating at mesophilic conditions (Vatten and Miljö i Väst AB, Varberg, Sweden). The inoculum was filtered through a 2 mm porosity sieve to remove unwanted particles, after which it was acclimated for five days in an incubator at 37 °C before use.

2.2. Substrate

Citrus wastes were taken from Brämhult juice AB (Borås, Sweden). The characteristics of the citrus waste used in this culture are shown in

Table 1
Characteristics of the citrus waste used in this experiment (Millati et al., 2018) with an update of D-Limonene content and pectin.

Components	Content (%)*
TS	23.99
VS	23.07
Water	79.07
Ash	0.88
Fat	0.23
Total Protein	1.28
Crude Fiber	3.28
Carbohydrate	18.54
Starch	0.62
Pectin	4.97
D-Limonene	3

* all data presented in dry basis (db) except TS. VS. and water.

Table 1.

2.3. Effect of initial O_2 presence, the addition of inhibitor for methanogens, and ISR on the production of VFAs

The anaerobic process was performed in order to examine the effect of different inoculum-substrate ratios, the effect of BES, and oxygen on the VFA production. This culture was studied with 16 different reactors in 8 different states (Table 1) in 120 mL of glass serum bottles as reactors with a working volume of 56 mL. The cultivations were conducted in duplicate for 14 days. When the inoculum/substrate ratio was 1:1, each bioreactor contained 1 g volatile solids (VS) of inoculum and 1 g VS of ground citrus waste. However, when the inoculum/substrate ratio is 1:3, each bioreactor contained 1 g VS of inoculum and 3 g VS of ground citrus waste. In this stage, the cultures in all reactors were adjusted to pH 6, and there was no pH adjustment for 14 days. The reactors were then closed and incubated under anaerobic conditions at 37.5 °C with 100 rpm. Each reactor put on a shaker for 14 days.

2.4. Effect of controlled pH conditions and ISR on the production of VFAs

The anaerobic digestion was performed in order to investigate the effect of different inoculum-substrate ratios, and the effect of pH on VFA production. This culture setup was conducted with six different states (Table 2). In this cultivation, 2 L glass bottle bioreactors was used as a batch digester with a total working volume of 1.5 L. Studies were continued for 22 days. When the inoculum/substrate ratio was 1:1, each bioreactor contained 25 g VS of free cells of inoculum and 25 g VS of ground citrus waste. However, when the inoculum/substrate ratio was 1:3, each bioreactor contained 25 g VS of free cells of inoculum and 75 g VS of ground citrus waste. In this cultivation, the experiment was carried out at three different pH of 4, 5, and 6. The pH was adjusted every two days to maintain the desired pH values constant. The pH adjustment was carried out by the addition of 2 M NaOH and/or 2 M HCl. The digesters were incubated under the anaerobic condition at 37.5 °C with 100 rpm.

2.5. Analytical methods

The moisture content, total solid (TS), volatile solid (VS), and volatile suspended solids (VSS) of substrates and inoculum were determined according to biomass analytical procedures (APHA-AWWA-WEF, 2005). Total Kjeldahl nitrogen (TKN) contents were measured using the Kjeldahl method, and the protein content was estimated by multiplying the TKN content with a factor of 6.25 (Nallathambi Gunaseelan, 2009). Total carbon was obtained by correcting the total dry weight carbon value for the ash content (Haug, 1993; Zhou et al., 2015). Fat content was determined using the Soxhlet extraction procedure (Carpenter, 2010). The pH and oxidation-reduction potential (ORP) were measured using pH- and ORP-meter (Mettler Toledo F20 FiveEasy, OH, USA) Samples from reactors were centrifuged (5 min at 20,000 × *g*), and supernatants were collected for the analyses of VFAs.

Table 2
Factors considered in the two-level full factorial design for investigation the effect of ISR, BES, and initial O_2.

Letter	Ratio	BES	Initial O_2 Presence
A	1:1	✓	×
B	1:1	✓	✓
C	1:1	×	×
D	1:1	×	✓
E	1:3	✓	×
F	1:3	✓	✓
G	1:3	×	×
H	1:3	×	✓

VFAs were measured using high-performance liquid chromatography (HPLC, Waters 2695, Waters Corporation, Milford, MA, USA) equipped with a biohydrogen-ion exchange column (Aminex HPX-87H, Bio-Rad, Hercules, CA, USA) operating at 60 °C. For the detection of VFAs, an UV absorbance detector (Walters 2487, Waters Corporation Milford, MA, USA), operating at 210 nm wavelength was used in series with a refractive index (RI) detector (Waters 2414, Waters Corporation Milford, MA, USA) operating at 60 °C. Prior to analysis, samples were first centrifuged at 20,000 × g for 10 min to exclude the natant from the analysis. The supernatant was then filtered using an HPLC certified syringe filter (GHP Acrodisc 13 mm with 0.2 μm GHP membrane) to remove remaining contaminant particles, which might hinder column separation.

Analysis of the dominant component of the citrus peel oil, i.e. D-limonene in the substrate and eluent was conducted using gas chromatography (GC) coupled to a mass spectroscopy detector (GC-MS Trace GC Ultra, ThemoScientific, Waltham, MA, USA). The GC-MS was equipped with a silica capillary column (ZB-5MS fused-silica with 30 m × 0.25 mm id × 0.25 μm film). Helium was used as the carrier gas with a flow rate of 1.2 mL/min. The temperature of the column was set at 50 °C for 2 min, and the temperature was increased to 120 °C with a rate of 4 °C/min. The injector temperature was set at 250 °C, and the detector temperature was set at 280 °C. Before analysis, the citrus essential oil was extracted from the sample by diluting 3 g of the sample with 10 mL of diethyl-ether and shaking overnight at 200 rpm.

The total volume of gas produced from 2 L CSTR (continuous stirred tank reactor) was measured by the tipping device in the Automatic Methane Potential Testing System (AMPTS, Bioprocess control AB, Lund, Sweden). The gas production, yield, and composition were determined based on a measurement using Gas Chromatography (Varian 450 GC, Palo Alto, CA, USA). A gas sample (100 μL) was withdrawn using a 250 μL gas-tight syringe equipped with a pressure lock (VICI, Baton Rouge, LA, USA). It was then injected and analyzed with the Varian 450 Gas Chromatography. The GC was equipped with a Wall Coated Open Tubular (WCOT, J & W Scientific GS-Gas Pro, bonded silica-based 30 m × 0.32 mm, Agilent Technologies, Santa Clara, CA, USA) capillary column, a thermal conductivity detector (TCD, Varian, Palo Alto, CA, USA) and Galaxie Chromatography Data System Single Instrument as the data recording software (v.1.9, Varian, Walnut Creek, CA, USA). Nitrogen was chosen as the carrier gas with flow rates of 2.0 mL/min passing the column and of 30 mL/min passing the detector. An injection split ratio was set at 5 with an injector temperature of 75 °C. Meanwhile, the oven, detector heater, and detector filament temperatures were set at 55, 120, and 200 °C, respectively. To ascertain peak retention time and calculate gas production, pure methane (CH_4), carbon dioxide (CO_2), and hydrogen (H_2) were used as the standard. The calculation was later done by comparing the peak area of the sample and the peak area of the standard, followed by conversion to the steady-state condition (273 K).

2.6. Experimental design and statistical analysis

A two-level three-factor full-factorial experiment design (2^k) was carried out to investigate the effects of ISR, initial O_2 presence, BES, and the interaction between effects towards VFA yield and methane production in the first culture (for culture effect of ISR, BES, and O_2). Meanwhile, the general multilevel full factorial design was carried out to investigate the effects of pH, ISR, and the interactions between two factors towards VFA yield and methane production (for culture effect of ISR and pH). The experimental design for each culture is presented in Tables 2 and 3. All experimental setups were carried out in duplicates.

In order to evaluate the influence of those factors, an analysis of variance (ANOVA) for the general multilevel factorial design was performed. The means of the significantly different main effects were compared at $p < 0.05$. The statistical analysis was performed using Minitab 19.

Table 3
Factors considered in the two-level full factorial design for investigation the effect of ISR and pH (controlled).

Letter	Ratio	pH
A	1:1	4
B	1:1	5
C	1:1	6
D	1:3	4
E	1:3	5
F	1:3	6

3. Results and discussions

Citrus waste, as a by-product of juice production, is a challenge for the environment. Its moisture content is too high to be a good feedstock for combustion and energy production. The presence of toxic compounds from peel oils such as D-limonene also makes it unattractive for composting and the biogas industries. Methanogens in anaerobic digestion (AD) are the most sensitive groups of the microorganisms. Hence, there is a possibility to stop the AD process before the methanogens and to produce VFAs instead. However, the question is if one can obtain a suitable yield of VFAs from citrus wastes. Therefore, in this work, the effects of different digestion conditions with no pH control during anaerobic digestion of citrus waste were investigated, and four response parameters, i.e., the yield of VFAs, the amount of biogas, the pH, and the oxidation-reduction potential (ORP) were studied. The most important results are summarized in Fig. 2. Based on the results of this first cultivations, in the second experimental setup, the effects of just ISR together with the effects of different controlled pH values were investigated by following up the same response parameters as within the first cultivations, and the results are summarized in Fig. 4.

3.1. Effect of initial O_2 presence, the addition of inhibitor for methanogens, and ISR on the production of VFAs

The conventional anaerobic digestion process can be shifted to produce VFAs instead of methane by stopping the degradation at the acidification stage. The conditions should be modified to be unsuitable for the VFA consumer, which is methanogens (Wainaina et al., 2019). In order to inhibit methanogens, several different approaches can be considered such as adding a chemical inhibitor, such as 2-bromoethanesulphonate (BES) (Zinder et al., 1984), or letting O_2 be present in the reactor (Wainaina et al., 2019), along with increasing the substrate loading (Wainaina et al., 2019). Citrus waste itself has been known to have an inhibitory effect on methane production in a biogas reactor (Fuertes, 2015; Kurniawan et al., 2018; Ruiz and Flotats, 2016; Ruiz Fuertes, 2015; Wikandari, 2014; Wikandari et al., 2015; Wikandari et al., 2014b). In this first cultivation, the effects of inoculum to substrate ratio (ISR), the addition of BES, and the presence of initial O_2 on the yield of VFAs obtained through batch digestion cultivations were determined and the results are presented in Fig. 1. All of these cultivations were carried out at an initial pH of 6, and then the pH remained uncontrolled during an incubation period of 14 days.

The change of pH values for 14 days for different inoculum-substrate ratios, BES, and oxygen conditions with initial pH 6 are shown in Fig. 1(a). In cases where the ratio of the inoculum-substrate is 1:1, the pH value has fallen below five on the second day, and on the 14th day, it has reached about 4.5 (Fig. 1(a)). On the other hand, when the ratio of the inoculum-substrate was 1:3, the pH value did not fall below five on the second day and decreased below 4.5 on the 14th day (Fig. 1(a)). In addition, in every case at initial pH 6, a sudden fall of pH occurred due to the production of VFAs on the second day and then continued with a closed curve. Cysneiros et al. (2012) showed in their study that pH dropped on the first day and then increased slightly and tended to stabilize at a particular value. Shahbaz et al. (2019) reported that the increase of VFA production caused the pH to decrease. In our study, the same inoculum-substrate ratios, presence or absence of BES and O_2 did not significantly affect the pH value.

ORP is an indicator of the net outcome of intracellular metabolism dominated by electron transfer and redox balance (Liu et al., 2013). The change of ORP (Oxidation-Reduction Potential) values for 14 days for different inoculum-substrate ratios, BES, and oxygen conditions are shown in Fig. 1(b). While ORP values were similar for all conditions at the beginning, ORP values were below 100 when the ratio of the inoculum to the substrate was 1:1 on the 14th day, but it was more than 130 on the 14th day in the case of 1:3. In both cases, the ORP values increased, but all ORP values tend to be stable even if there are minor changes from the 8th day.

In this cultivation, two different inoculum/substrate ratios have been studied to be 1:1 and 1:3 in order to investigate how VFA production is affected when the amount of substrate is increased. As shown in Fig. 2(a–h). When the amount of substrate is increased, the yield of VFAs decreased. While the substrate to the inoculum ratio was 1:1, the amount of acetate and butyrate increased almost the same until the 14th day. When the substrate to inoculum ratio increased (from 1:1 to 1:3), the amount of acetate increased more than the amount of butyrate. Similar results were obtained in the study by Xu et al. (2012), BES was added to prevent VFA consumption by inhibiting the methanogenesis under anaerobic conditions. Fig. 2 indicates both in the presence and absence of BES, the profile of the concentration of total VFA, as well as the individual VFA, including acetic acid, propionic acid, isobutyric acid, butyric acid, isovaleric acid, valeric acid, and caproic acid at pH 6. BES is a methanogen inhibitor. Since the addition of BES in anaerobic conditions prevents methanogen production, VFA production is possible. However, as it is observed in Fig. 2(a, b, e, f with BES vs. c, d, g, h with no BES), the addition of the BES did not make

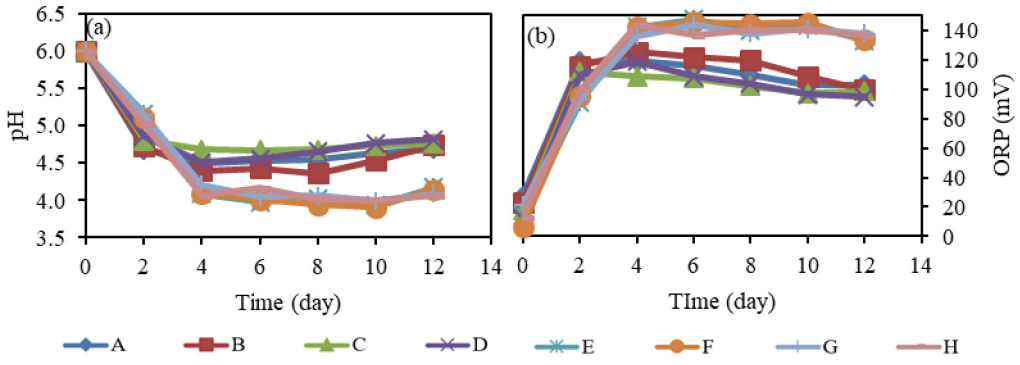


Fig. 1. Change of pH and ORP (oxidation-reduction potential) at different inoculum to substrate ratios (ISR), presence and absence of BES and oxygen. The letters correspond to different conditions as mentioned in Table 2.

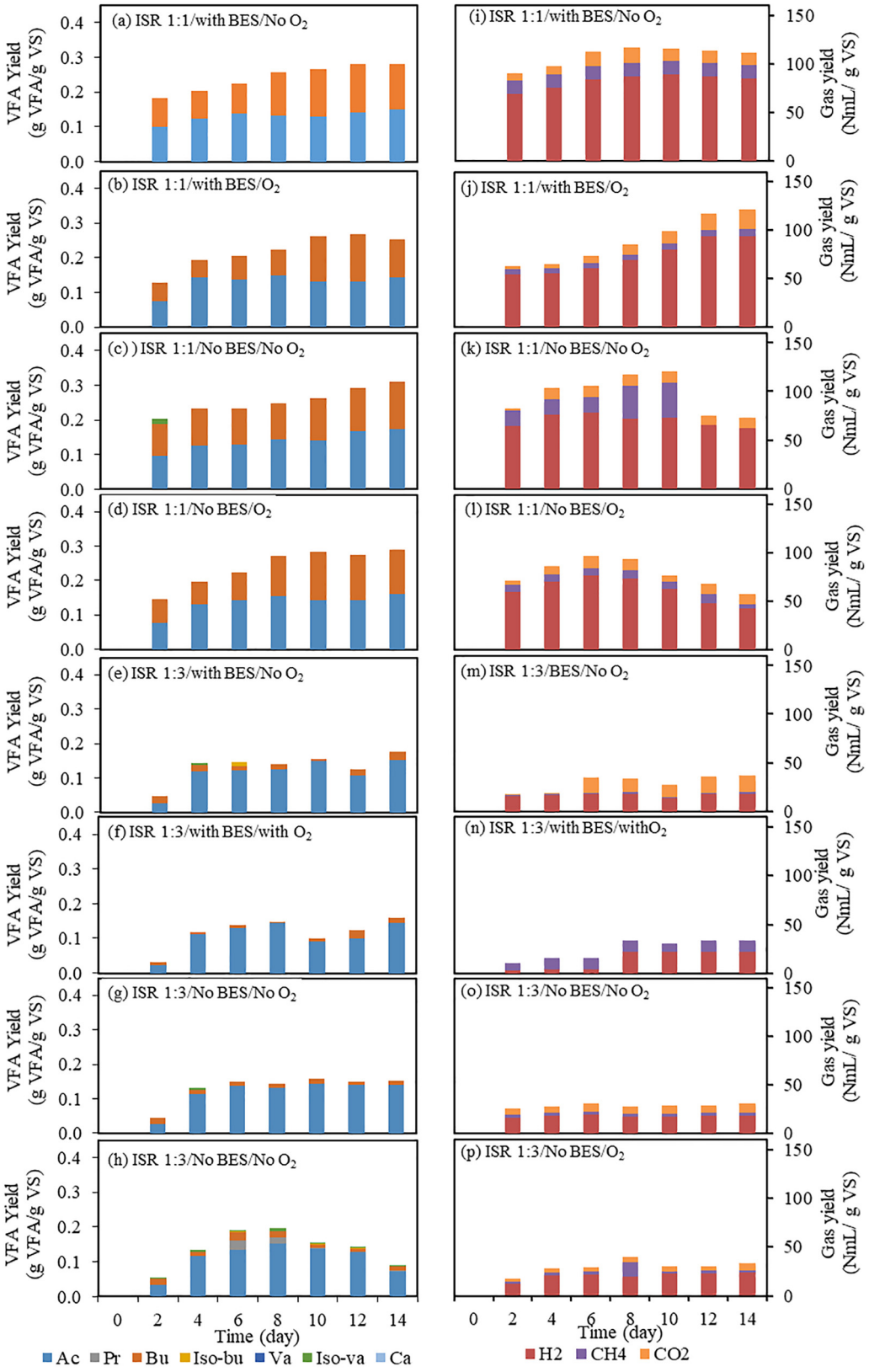


Fig. 2. VFA (volatile fatty acid) and biogas yield during batch anaerobic digestion of citrus waste at pH 6 using different ISR, with or without the addition of BES, and with or without the presence of initial O_2. The abbreviations of the VFAs are: Ac: acetic acid, Pr: propionic acid, Bu: butyric acid, Iso-bu: Iso-butyric acid, Va: Valeric acid, Iso-va: Iso-valeric acid, Ca: Caproic acid.

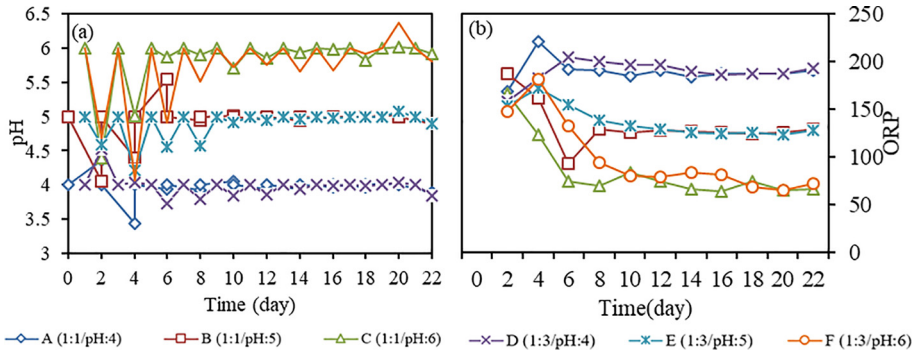


Fig. 3. Change of pH and ORP (oxidation-reduction potential) at different ISR (inoculum to substrate ratios) and pH. The letters correspond to different conditions as mentioned in Table 3.

significant changes in the amount of VFA.

In order to investigate the effect of fully anaerobic conditions (without O_2 presence) on VFA production, the headspace was flushed with a mixture of % 20 CO_2 and% 80 N_2. The initial presence of oxygen did not have a significant effect (Fig. 2 a, c, e, g vs. b, d, f, h) on the VFA yield. This fact can be beneficial in the application on a larger scale. Unlike a methane-based anaerobic digester, this VFA-based reactor is not as sensitive to oxygen. Since this study was performed under anaerobic conditions and the last product of the anaerobic digestion is normally biogas, the gas formation was evaluated in this study. The biogas production and composition for 14 days for different inoculum-substrate ratios, BES, and oxygen conditions are shown in Fig. 2(i–p). When comparing Fig. 4(e) and (g), CH_4 and CO_2 contents appear to be higher in the non-BES reactor (Fig. 4(g)). Compared to Fig. 4(a) and (c), hydrogen production, an intermediate product in anaerobic processes such as VFA, appears to be higher in the BES-containing reactor A (Fig. 4(a)). In comparison with Fig. 4(a) and (b), oxygen did not have a significant impact on biogas production.

Among the three factors examined in this culture, inoculum to substrate ratio has been shown to have a significant impact on VFA yield. Higher substrate loading causes a decrease in VFA yield. Not only affecting methanogens, citrus peel oil containing in the citrus waste, which predominantly consists of D-Limonene, might also inhibit the acidification of VFAs (Kurniawan et al., 2018; Lukitawesa et al., 2018). The presence of initial O_2 and the addition of inhibitor for methanogens mostly affect biogas production and do not significantly affect the VFA yield. Since the presence of O_2 did not negatively affect the VFA production, an acidification reactor which is treating citrus waste may not have to be as strictly anaerobic as a methane production reactor. This fact can simplify the maintenance of the reactor, during which it sometimes needs to be opened.

3.2. Effect of different controlled pH on VFA production

In the second culture, the effect of pH control was investigated together with the inoculum substrate ratio. Four parameters were observed, i.e., volatile fatty acid (VFA) production and volatile fatty acid (VFA) yield, gas production, pH, and ORP. The cultivations were carried out by increasing pH (4, 5 and 6) and inoculum substrate ratio (1:1, 1:3). Since pH and ORP are essential parameters for system stabilization in biological systems, pH and ORP values have been recorded in this culture for 22 days.

In this cultivation, the pH at three different pH values was controlled during 22 days, where the pH was adjusted back to the starting pH every second days. The change of pH values for different inoculum-substrate ratios and different pH controlled or uncontrolled were measured during the experiment. In all conditions (Fig. 5(a) and (b)), pH values showed significant changes during the first six days. However, from the 8th day onwards, the changes were more stable and closer to the constant level. Moreover, in general, a sudden fall occurred due to the production of VFAs on the second day and then continued with a closed curve. Similarly, Kim et al. (2003) found that at the beginning of the acidification of food waste, the pH dropped steeply and stabilized afterward.

The change of ORP values for different inoculum-substrate ratios and different pH values is presented in Fig. 3(b). The ORP values for all conditions were fluctuated during the first eight days. After the 8th day, the ORP become more stable. ORP decreased, and VFA accumulated slowed after day 8. A similar trend was previously reported by Yin et al. (2016). When the inoculum-substrate ratios were 1:1 and 1:3, and the pH was 4, the ORP value was around 190 at the end of the 2 s day, while at pH 5 it was about 125 and pH 6 it was about 70.

In order to determine the effect of pH change on the production of VFAs in this culture, three different pH values (4, 5, and 6) were examined. An extremely acidic (less than 3) or extremely alkaline pH (more than 12) can inhibit the acidogens (Wainaina et al., 2019). Therefore, this study was carried out with three different pH values of 4, 5 and 6. VFA yield tended to decrease when pH was increased (Fig. 4(a–f) and Table 5). In both cases (1:1 and 1:3), the highest VFA production and yield were at pH 6. Jiang et al. (2013) reported also a similar study with three different pH values (5, 6 and 7) and reached the highest VFA production at pH 6. At pH 4, while one of the dominant species was propionate, the propionate value decreased as the pH gets higher, and it was undetectable. When the pH increased, acetate and butyrate yields increased, as observed in Fig. 4(a and c), while no caproate value was present at pH 4. Caproate was produced at pH 5 and 6, as shown in Fig. 4(b, c, e, and f).

In this second culture, it was investigated how the ratio of inoculum substrate with different pH values affects the production of VFAs. The VFA yield decreased as the ratio of inoculum substrate changed from 1:1 to 1:3 (Table 4). Since this study was performed under anaerobic conditions and the last product of the anaerobic digestion was biogas. Gas measurements were made in this study. The biogas production and composition for 22 days for different inoculum-substrate ratios and different pH values are shown in Fig. 4(g–l). The incubation time was only 22 days because, in our preliminary study, there was no increase in VFA production afterward (data not shown). Studies by Zhou et al. (2016) showed that the performance of anaerobic digestion was strongly dependent on pH value. They worked with three different pH and achieved the highest amount of biogas 16.6 mL at pH 7.

The biogas yield increased as the pH increased (Fig. 4(i–k)). When pH was constant, and the biogas production decreased as the inoculum-substrate ratio increased. In addition, biogas production decreased as pH increased compared to Fig. 4(l–n).

3.3. The importance of controlling pH on VFA production during digestion

One of the crucial factors to achieving a high VFA yield is pH. The

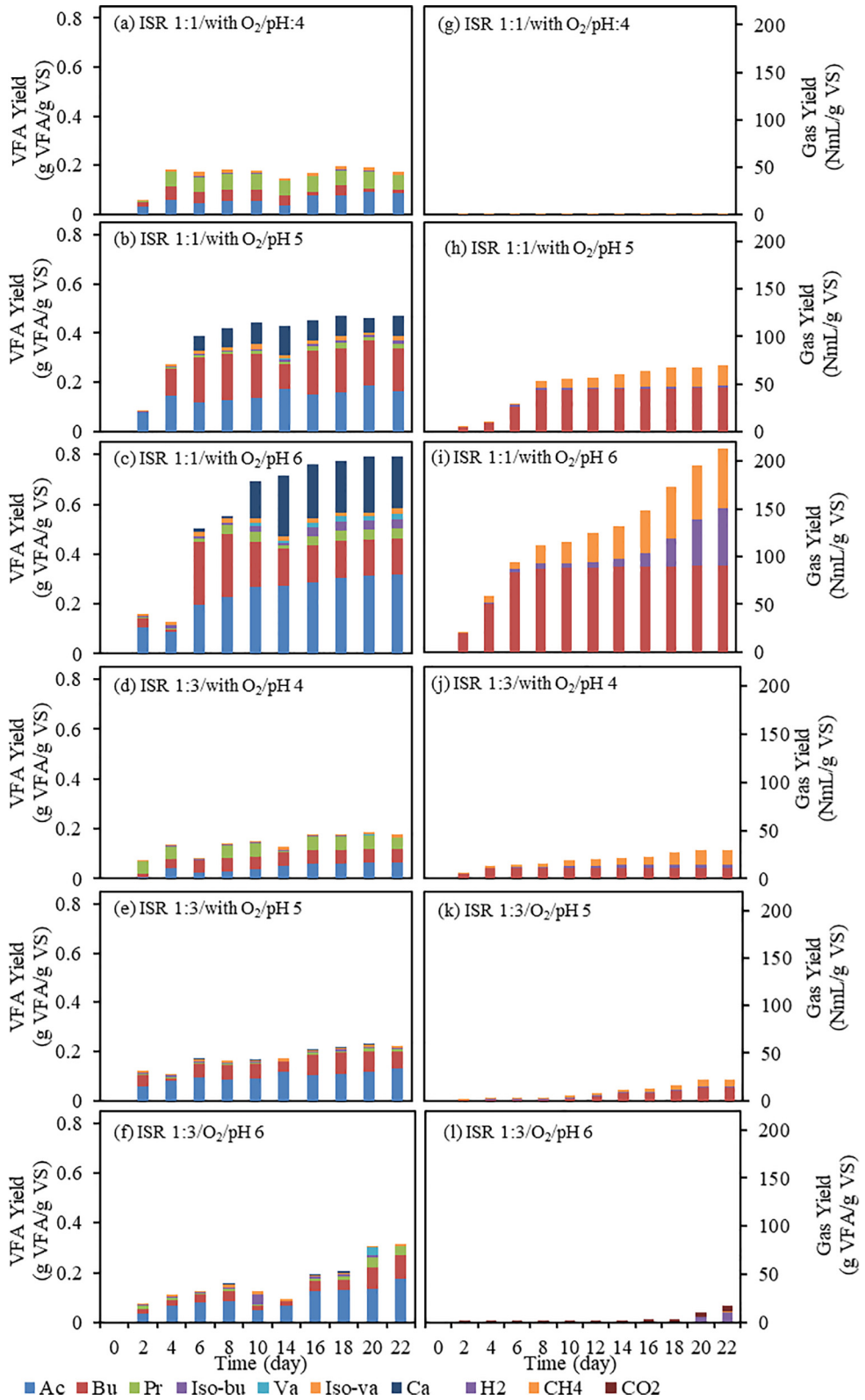


Fig. 4. VFA (volatile fatty acid) and biogas yield during batch fermentation of citrus waste at different controlled pH and ISR (inoculum to substrate ratios). The abbreviations of the acids are similar to Fig. 2.

pH 6 was shown to be a suitable condition to produce high VFA yield compared to lower pH (Fig. 4). However, adjusting the initial pH was not enough to achieve a high VFA yield. As shown in Fig. 5(a), initial pH adjustment at pH 6 without pH controlling during digestion resulted in lower VFA yield (0.269 g VFA/g VS) compared to the reactor with pH control at 6 (Fig. 5(b)) which had a VFA yield of 0.793 g VFA/g VS. Meanwhile, controlling the pH at 6 at higher substrate loading (ISR 1:3) increased the VFA yield from 0.161 g VFA/g VS into 0.314 g VFA/g VS Fig. 5(c and d). Controlling the pH throughout digestion is necessary because VFAs accumulation can cause pH to drop. When the pH

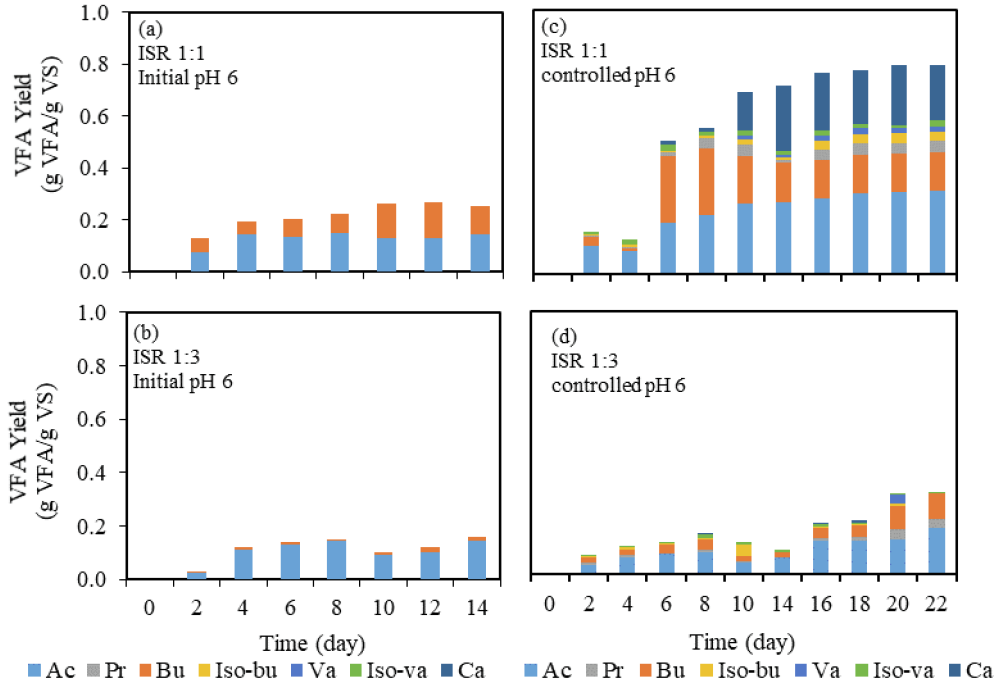


Fig. 5. VFA (inoculum to substrate ratios) yield during batch fermentation of citrus waste at initially adjusted to pH 6 but uncontrolled (a and b) and controlled at pH 6 through the experiments (c and d). The abbreviations of the individual acids are similar to Fig. 2.

dropped until under the pKa value of VFAs, most of the VFA will be in undissociated form. These undissociated acids can easily pass across the cell membrane and dissociate inside the cell, causing the energy generated in the cell to be directed towards cell maintenance, which eventually hinders the microbial productivity (van der Wielen et al., 2000).

4. Conclusions

Citrus waste can be converted into VFAs through controlled anaerobic digestion at low substrate loading. The most significant factor in this study is inoculum to substrate ratio and controlling pH. The highest yield of VFAs of 0.793 g VFA/VS was achieved by controlling the pH at 6 and low substrate loading (ISR 1:1) with the presence of initial O_2. Higher substrate loading resulted in the VFA yield to decrease. In practical application, controlling the pH is most likely the most important factor determining VFA yield. Besides, as oxygen does not have a significant effect on VFA yield, the presence of small amount of oxygen or opening the reactor for maintenance does not harm the digestion process.

CRediT authorship contribution statement

Bahriye Eryildiz: Data curation, Visualization, Writing - original draft, Investigation, Funding acquisition. **Lukitawesa:** Visualization, Conceptualization, Methodology, Writing - original draft, Writing - review & editing, Investigation, Funding acquisition, Formal analysis. **Mohammad J. Taherzadeh:** Project administration, Methodology, Validation, Resources, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 4
VFA yield during batch fermentation at different inoculum to substrate ratio, with and without the presence of BES, and with and without the presence of O_2. The abbreviations of the acids are similar to Fig. 2.

Condition	Ac	Pr	Bu	Iso-bu	Va	Iso-va	Ca	Total VFA Yield (g VFA/ VS_{added})	Day	Total VFA Yield (g COD_{VFA}/ COD_{input})*	D-Limonene concentration in the reactor (g/L)**
(a) 1:1/BES/No O_2	0.151	0	0.129	0	0	0	0	0.280	14	0.227	0.557
(b) 1:1/BES/O_2	0.131	0	0.138	0	0	0	0	0.269	12	0.222	0.557
(c) 1:1/No BES/No O_2	0.173	0	0.138	0	0	0	0	0.312	14	0.251	0.557
(d) 1:1/No BES/O_2	0.162	0	0.127	0	0	0	0	0.289	14	0.232	0.557
(e) 1:3/BES/No O_2	0.152	0	0.023	0	0	0	0	0.175	14	0.126	1.670
(f) 1:3/BES/O_2	0.144	0	0.016	0	0	0	0	0.161	14	0.114	1.670
(g) 1:3/No BES/No O_2	0.139	0	0.013	0	0	0	0	0.152	14	0.107	1.670
(h) 1:3/No BES/O_2	0.072	0.003	0.011	0.002	0	0.001	0	0.089	14	0.066	1.670

* Calculated based on conversion factor of VFA to COD value.
** D-Limonene concentrations inside each reactor was calculated based on the concentration in the substrate as shown in Table 1.

Table 5
VFA yield during anaerobic digestion of citrus waste at different controlled pH and ISR. The abbreviations of the acids are similar to Fig. 2.

Condition	Ac	Pr	Bu	Iso-bu	Va	Iso-va	Ca	Total VFA Yield (g VFA/ VS_{added})	Day	Total VFA Yield (g COD_{VFA}/ COD_{input})*	D-Limonene concentration in the reactor (g/L)**
(a) 1:1/pH:4	0.079	0.062	0.039	0.002	0	0.014	0	0.196	18	0.184	0.0005
(b) 1:1/pH:5	0.163	0.016	0.177	0.014	0	0.020	0.082	0.471	22	0.456	0.0005
(c) 1:1/pH:6	0.317	0.042	0.145	0.036	0.019	0.024	0.209	0.793	22	0.796	0.0005
(d) 1:3/pH:4	0.064	0.053	0.054	0.004	0	0.009	0	0.185	20	0.174	0.0016
(e) 1:3/pH:5	0.132	0.009	0.068	0.003	0.001	0.010	0	0.223	22	0.183	0.0016
(f) 1:3/pH:6	0.175	0.038	0.093	0.001	0	0.007	0	0.314	22	0.263	0.0016

* Calculated based on conversion factor of VFA to COD value.
** D-Limonene concentrations inside each reactor was calculated based on the concentration in the substrate as shown in Table 1.

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References

APHA-AWWA-WEF, 2005. Standard Methods for the Examination of Water and Wastewater, 21st ed. American Public Health Association (APHA), American Water Works Association (AWWA), Water Environment Federation (WEF), Washington DC.
C. Carpenter, Determination of Fat Content, Food Analysis Laboratory Manual: Food Science Texts Series, 2010, pp. 29–37.
Chidthaisong, A., Conrad, R., 2000. Specificity of chloroform, 2-bromoethanesulfonate and fluoroacetate to inhibit methanogenesis and other anaerobic processes in anoxic rice field soil. Soil Biol. Biochem. 32 (7), 977–988.
Clarke, S., 2008. Chapter 7 – Composition of essential oils and other materials. In: Clarke, S. (Ed.), Essential Chemistry for Aromatherapy, Second ed. Churchill, Livingstone, Edinburgh, pp. 123–229.
Cysneiros, D., Banks, C.J., Heaven, S., Karatzas, K.A.G., 2012. The effect of pH control and 'hydraulic flush' on hydrolysis and volatile fatty acids (VFA) production and profile in anaerobic leach bed reactors digesting a high solids content substrate. Bioresour. Technol. 123, 263–271.
Dahiya, S., Sarkar, O., Swamy, Y.V., Mohan, S.V., 2015. Bioresource technology acidogenic fermentation of food waste for volatile fatty acid production with co-generation of biohydrogen. Bioresour. Technol. 182, 103–113.
Debruyn, J., Hilborn, D., 2007. Anaerobic digestion basics. Small 2007, 1–6.
FAO, 2017. Citrus Fruit Fresh and Processed Statistical. Food and Agriculture Organization of the United Nations.
Fuertes, B.R., 2015. Effect of limonene on anaerobic digestion of citrus waste and pretreatments for its improvement Vol. PhD. Universitat Politecnica de Valencia, Valencia, pp. 216.
Haug, R.T., 1993. In the Practical Handbook of Compost Engineering. Taylor & Francis, pp. 247–257.
Jankowska, E., Chwialkowska, J., Stodolny, M., Oleskowicz-popiel, P., 2017. Volatile fatty acids production during mixed culture fermentation – the impact of substrate complexity and pH. Chem. Eng. J. 326, 901–910.
Jiang, J., Zhang, Y., Li, K., Wang, Q., Gong, C., Li, M., 2013. Volatile fatty acids production from food waste: effects of pH, temperature, and organic loading rate. Bioresour. Technol. 143, 525–530.
Khan, M.A., Ngo, H.H., Guo, W.S., Liu, Y., Nghiem, L.D., Hai, F.I., Deng, L.J., Wang, J., Wu, Y., 2016. Optimization of process parameters for production of volatile fatty acid, biohydrogen and methane from anaerobic digestion. Bioresour. Technol. 219, 738–748.
Kim, M., Gomec, C.Y., Ahn, Y., Speece, R.E., 2003. Hydrolysis and acidogenesis of particulate organic material in mesophilic and thermophilic anaerobic digestion. Env. Technol. (UK) 24, 1183–1190.
Kurniawan, T., Lukitawesa, Hanifah, I., Wikandari, R., Millati, R., Taherzadeh, M.J., Niklasson, C., 2018. Semi-continuous reverse membrane bioreactor in two-stage anaerobic digestion of citrus waste. Mater. (Basel, Switzerland) 11 (8), 1341.
Li, Y., Hua, D., Zhang, J., Zhao, Y., Xu, H., Liang, X., Zhang, X., 2013. Volatile fatty acids distribution during acidogenesis of algal residues with pH control. World J. Microbiol. Biotechnol. 29 (6), 1067–1073.
Li, Y., Jin, Y., Borrion, A., Li, H., Li, J., 2017. Effects of organic composition on mesophilic anaerobic digestion of food waste. Bioresour. Technol. 244, 213–224.
Liu, C.-G., Xue, C., Lin, Y.-H., Bai, F.-W., 2013. Redox potential control and applications in microaerobic and anaerobic fermentations. Biotechnol. Adv. 31 (2), 257–265.
Lukitawesa, Patinvoh, R.J., Millati, R., Sárvári-Horváth, I., Taherzadeh, M.J., 2020. Factors influencing volatile fatty acids production from food wastes via anaerobic digestion. Bioengineered 11 (1), 39–52.
Lukitawesa, Wikandari, R., Millati, R., Taherzadeh, M.J., Niklasson, C., 2018. Effect of effluent recirculation on biogas production using two-stage anaerobic digestion of citrus waste. Molecules 23 (12).
Marín, F.R., Soler-Rivas, C., Benavente-García, O., Castillo, J., Pérez-Alvarez, J.A., 2007. By-products from different citrus processes as a source of customized functional fibres. Food Chem. 100, 736–741.
D.I. Massé, G. Talbot, Y. Gilbert, A scientific review of the agronomic, environmental and social benefits of anaerobic digestion, in: Anaerobic Digestion: Processes, Products and Applications, pp. 109–131.
Millati, R., Lukitawesa, Permanasari, E.D., Sari, K.W., Cahyanto, M.N., Niklasson, C., Taherzadeh, M.J., 2018. Anaerobic digestion of citrus waste using two-stage membrane bioreactor. IOP Conference Series: Materials Science and Engineering. IOP Publishing.
Mohan, S.V., Babu, V.L., Sarma, P.N., 2008. Effect of various pretreatment methods on anaerobic mixed microflora to enhance biohydrogen production utilizing dairy wastewater as substrate. Bioresour. Technol. 99 (1), 59–67.
Nallathambi Gunaseelan, V., 2009. Biomass estimates, characteristics, biochemical methane potential, kinetics and energy flow from Jatropha curcus on dry lands. Biomass Bioenergy 33 (4), 589–596.
Ruiz, B., de Benito, A., Rivera, J.D., Flotats, X., 2016. Assessment of different pre-treatment methods for the removal of limonene in citrus waste and their effect on methane potential and methane production rate. Waste Manage. Res. 34 (12), 1249–1257.
Ruiz, B., Flotats, X., 2014. Citrus essential oils and their influence on the anaerobic digestion process: an overview. Waste Manage. 34, 2063–2079.
Ruiz, B., Flotats, X., 2016. Effect of limonene on batch anaerobic digestion of citrus peel waste. Biochem. Eng. J. 109, 9–18.
B. Ruiz Fuertes, Effect of limonene on anaerobic digestion of citrus waste and pretreatments for its improvement, 2015.
Shahbaz, M., Ammar, M., Zou, D., Korai, R.M., Li, X.J., 2019. An insight into the anaerobic co-digestion of municipal solid waste and food waste: influence of co-substrate mixture ratio and substrate to inoculum ratio on biogas production. Appl. Biochem. Biotechnol. 187, 1356–1370.
Stein, U.H., Wimmer, B., Ortner, M., Fuchs, W., Bochmann, G., 2017. Maximizing the production of butyric acid from food waste as a precursor for ABE-fermentation. Sci. Total Environ. 598, 993–1000.
Trad, Z., Akimbomi, J., Vial, C., Larroche, C., Taherzadeh, M.J., Fontaine, J.-P., 2015. Bioresource technology development of a submerged anaerobic membrane bioreactor for concurrent extraction of volatile fatty acids and biohydrogen production. Bioresour. Technol. 196, 290–300.
van der Wielen, P.W.J.J., Biesterveld, S., Notermans, S., Hofstra, H., Urlings, B.A.P., van Knapen, F., 2000. Role of volatile fatty acids in development of the cecal microflora in broiler chickens during growth. Appl. Environ. Microbiol. 66 (6), 2536–2540.
Wainaina, S., Lukitawesa, Kumar Awasthi, M., Taherzadeh, M.J., 2019. Bioengineering of anaerobic digestion for volatile fatty acids, hydrogen or methane production: a critical review. Bioengineered 10 (1), 437–458.
Wang, K., Yin, J., Shen, D., Li, N., 2014. Anaerobic digestion of food waste for volatile fatty acids (VFAs) production with different types of inoculum: Effect of pH. Bioresour. Technol. 161, 395–401.
Wikandari, R., 2014. Effect of fruit flavors on anaerobic digestion: inhibitions and solutions. Skrifter från Högskolan i Borås. University of Borås, School of Engineering.
Wikandari, R., Millati, R., Cahyanto, M.N., Taherzadeh, M.J., 2014a. Biogas production from citrus waste by membrane bioreactor. Membranes 4, 596–607.
Wikandari, R., Nguyen, H., Millati, R., Niklasson, C., Taherzadeh, M.J., 2015. Improvement of biogas production from orange peel waste by leaching of limonene. BioMed Res. Int.
Wikandari, R., Youngsukkasem, S., Millati, R., Taherzadeh, M.J., 2014b. Performance of semi-continuous membrane bioreactor in biogas production from toxic feedstock containing d-limonene. Bioresour. Technol. 170, 350–355.
Wu, F., Jin, Y., Xu, X., Yang, N., 2017. Electrofluidic pretreatment for enhancing essential oil extraction from citrus fruit peel waste. J. Cleaner Prod. 159, 85–94.
Xie, L., Liu, H., Chen, Y.G., Zhou, Q., 2014. pH-adjustment strategy for volatile fatty acid production from high-strength wastewater for biological nutrient removal. Water Sci. Technol. 69 (10), 2043–2051.
Xu, F., Li, Y., Ge, X., Yang, L., Li, Y., 2018. Anaerobic digestion of food waste – challenges and opportunities. Bioresour. Technol. 247, 1047–1058.
Xu, S.Y., Karthikeyan, O.P., Selvam, A., Wong, J.W.C., 2012. Effect of inoculum to substrate ratio on the hydrolysis and acidification of food waste in leach bed reactor. Bioresour. Technol. 126, 425–430.
Yin, J., Yu, X., Zhang, Y., Shen, D., Wang, M., Long, Y., Chen, T., 2016. Enhancement of acidogenic fermentation for volatile fatty acid production from food waste: effect of redox potential and inoculum. Bioresour. Technol. 216, 996–1003.
Zhou, C., Liu, Z., Huang, Z.L., Dong, M., Yu, X.L., Ning, P., 2015. A new strategy for co-composting dairy manure with rice straw: addition of different inocula at three stages of composting. Waste Manage. 40, 38–43.

Zhou, J., Zhang, R., Liu, F., Yong, X., Wu, X., Zheng, T., Jiang, M., Jia, H., 2016. Biogas production and microbial community shift through neutral pH control during the anaerobic digestion of pig manure. Bioresour. Technol. 217, 44–49.
Zhou, M., Yan, B., Wong, J.W.C., Zhang, Y., 2018. Enhanced volatile fatty acids production from anaerobic fermentation of food waste: a mini-review focusing on acidogenic metabolic pathways. Bioresour. Technol. 248, 68–78.
Zinder, S.H., Anguish, T., Cardwell, S.C., 1984. Selective inhibition by 2-bromoethanesulfonate of methanogenesis from acetate in a thermophilic anaerobic digestor. Appl. Environ. Microbiol. 47 (6), 1343–1345.